First: System Vision document

Smallholder farmers face struggles while participating in carbon credit programs due to the difficulty of the process and high costs because their farms are too small and less qualified to reduce carbon dioxide emissions and greenhouse gases. In addition to, companies and factories around the world buy carbon credits to offset their emissions and struggle from not knowing how to compensate the emissions coming out from them.

The **Carbon CV (carbon credit verification system)** is a digital platform that helps farmers earn money from carbon credits. It groups small farms into large projects, monitors their activities, and verifies their results so carbon credit can be issued and sold to companies. Our system brings farmers, field agents, land officers, verifiers, and companies (stakeholders) that are in need to buy carbon credits to reduce their carbon dioxide emissions and greenhouse gases together. It aims to make the verification procedures more transparent, easier, and more organized.

Our system capabilities:

- Gathering information about farmers, field agents, plots, land officers and companies
- Registering plots and managing the plot boundaries, areas and practices
- Monitoring farmers' activities by capturing satellite images
- Detecting violations as burning, land clearing, and sending SMS warnings to the farmers. Also checking farmers' uploaded photos as evidence (photos with location and timestamp).
- Grouping plots (small farms) into clusters so it could meet the system qualifications (at least 500 hectares)
- Handling verification process by inspecting random samples from each cluster (at least 5%) and upon these samples our team will decide if this cluster is approved or not
- Issuing, tracking carbon credits and Save carbon credit information
- Strengthening the connections between stakeholders

Our main aim is supporting eco-friendly farms, helping farmers earn extra income, and simplify the verification process on them and making small farms more reachable for companies to purchase their need of carbon credits from them. The challenge is handling grouping, monitoring, and verifying processes before any carbon credit issued and sold.